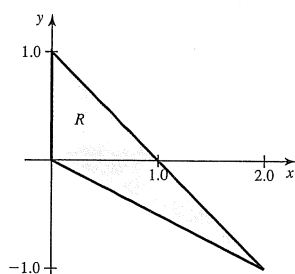
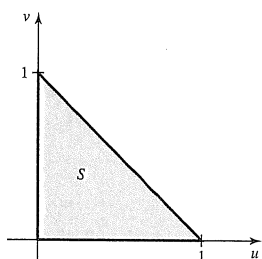


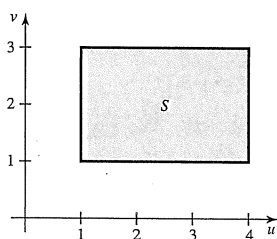
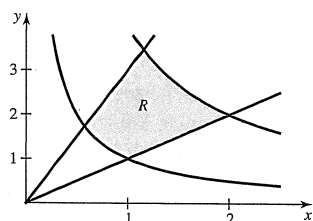
75. a.



b. $0 \leq u \leq 1, 0 \leq v \leq 1 - u$ c. $J(u, v) = 2$ d. $\frac{256\sqrt{2}}{945}$



77. 42

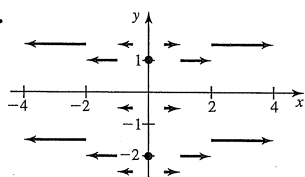
79. $-\frac{7}{16}$

CHAPTER 15

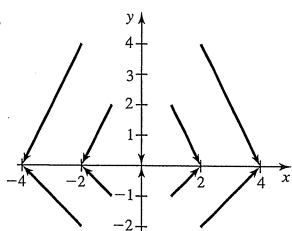
Section 15.1 Exercises, pp. 957–959

1. $\mathbf{F} = \langle f, g, h \rangle$ evaluated at (x, y, z) is the velocity vector of an air particle at (x, y, z) at a fixed point in time. 3. At selected points (a, b) , plot the vector $\langle f(a, b), g(a, b) \rangle$. 5. It shows the direction in which the temperature increases the fastest and the amount of increase.

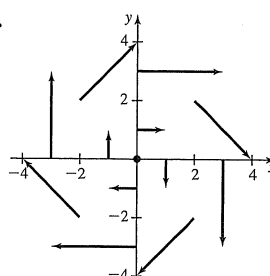
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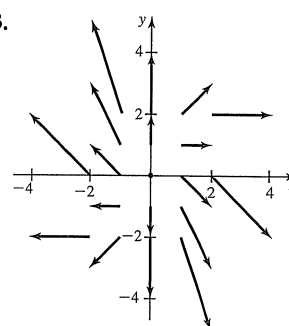
9.



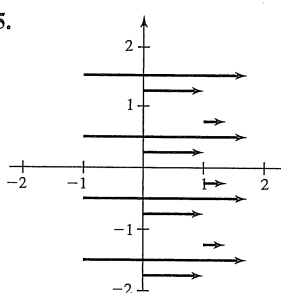
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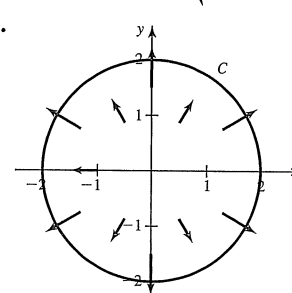
13.



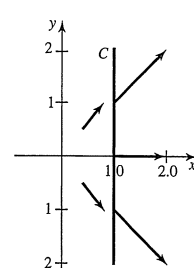
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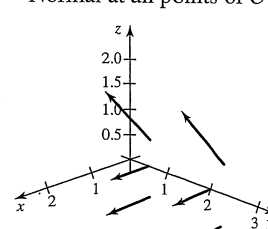
17.

Normal at all points of C

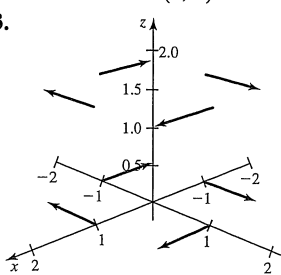
19.

Normal to C at $(1, 0)$

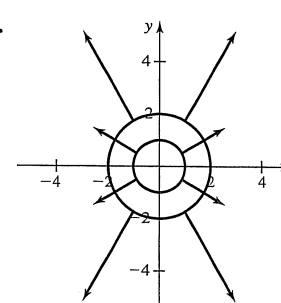
21.



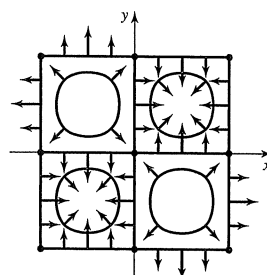
23.



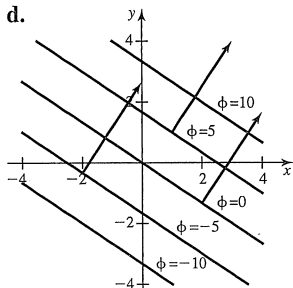
25.

 $\nabla\phi(x, y) = 2\langle x, y \rangle$

27.

 $\nabla\phi(x, y) = \langle \cos x \sin y, \sin x \cos y \rangle$ 29. $\nabla\phi(x, y, z) = \langle x, y, z \rangle = \mathbf{r}$ 31. $\nabla\phi(x, y, z) = -(x^2 + y^2 + z^2)^{-3/2} \langle x, y, z \rangle = -\frac{\mathbf{r}}{|\mathbf{r}|^3}$

33. a. $\nabla\varphi(x, y) = \langle 2, 3 \rangle$ b. $y' = -2/3, \langle 1, -2/3 \rangle \cdot \nabla\varphi(1, 1) = 0$
 c. $y' = -2/3, \langle 1, -2/3 \rangle \cdot \nabla\varphi(x, y) = 0$ d.

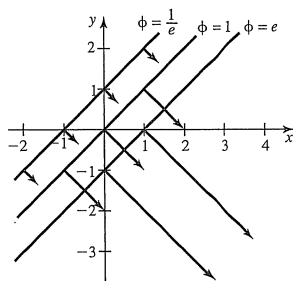


35. a. $\nabla\varphi(x, y) = \langle e^{x-y}, -e^{x-y} \rangle = e^{x-y} \langle 1, -1 \rangle$

b. $y' = 1, \langle 1, 1 \rangle \cdot \nabla\varphi(1, 1) = 0$

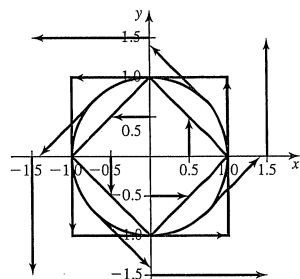
c. $y' = 1, \langle 1, 1 \rangle \cdot \nabla\varphi(x, y) = 0$

d.



37. a. True b. False c. True

39.



- a. For S and D , the vectors with maximum magnitude occur at the vertices; on C all vectors on the boundary have the same maximum magnitude ($|\mathbf{F}| = 1$). b. For S and D the field is directed out of the region on line segments between any vertex and the midpoint of the boundary line when proceeding in a counterclockwise direction; on C the vector field is tangent to the boundary curve everywhere.

41. $\mathbf{F} = \langle -y, x \rangle$ or $\mathbf{F} = \langle -1, 1 \rangle$

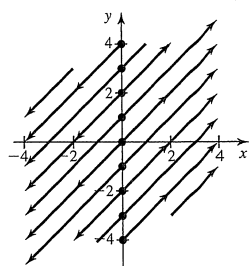
43. $\mathbf{F}(x, y) = \frac{\langle x, y \rangle}{\sqrt{x^2 + y^2}} = \frac{\mathbf{r}}{|\mathbf{r}|}, \mathbf{F}(0, 0) = 0$

45. a. $E = \frac{c}{x^2 + y^2} \langle x, y \rangle$ b. $E = \frac{c}{|\mathbf{r}|^2} \mathbf{r} = \frac{c}{|\mathbf{r}|} \frac{\mathbf{r}}{|\mathbf{r}|}$

c. Hint: The equipotential curves are circles centered at the origin.

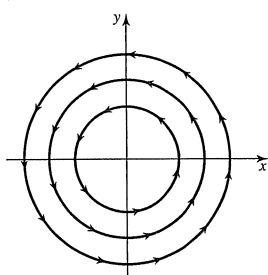
47. The tangent vector $\mathbf{v} = \langle 1, y'(x) \rangle$ to $y(x)$ is parallel to $\mathbf{F}$.

49.



$y = x + C$

51.



$x^2 + y^2 = C$

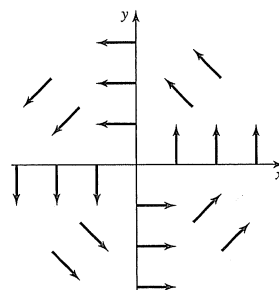
53. For $\theta = 0$: $\mathbf{u}_r = \mathbf{i}$ and $\mathbf{u}_\theta = \mathbf{j}$

for $\theta = \frac{\pi}{2}$: $\mathbf{u}_r = \mathbf{j}$ and $\mathbf{u}_\theta = -\mathbf{i}$

for $\theta = \pi$: $\mathbf{u}_r = -\mathbf{i}$ and $\mathbf{u}_\theta = -\mathbf{j}$

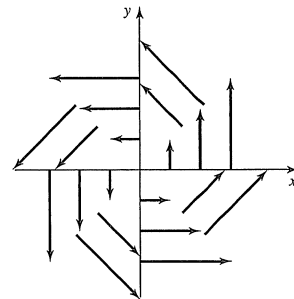
for $\theta = \frac{3\pi}{2}$: $\mathbf{u}_r = -\mathbf{j}$ and $\mathbf{u}_\theta = \mathbf{i}$

55. 57.



$\mathbf{F} = \frac{1}{\sqrt{x^2 + y^2}} \langle -y, x \rangle$

$\mathbf{F} = r \mathbf{u}_\theta$



Section 15.2 Exercises, pp. 973–977

1. A line integral is taken along a curve, an ordinary single-variable integral is taken along an interval. 3. $\sqrt{1 + 4t^2}$ 5. The integrand of the alternate form is a dot product of $\mathbf{F}$ and $\mathbf{T}$ ds . 7. Take the line integral of $\mathbf{F} \cdot \mathbf{T}$ along the curve with arc length as the parameter.

9. Take the line integral of $\mathbf{F} \cdot \mathbf{n}$ along the curve with arc length as the parameter, where $\mathbf{n}$ is the outward normal vector of the curve.

11. 0 13. $-\frac{32}{3}$ 15. a. $\mathbf{r}(t) = \langle 4 \cos t, 4 \sin t \rangle, 0 \leq t \leq 2\pi$

b. $|\mathbf{r}'(t)| = 4$ c. 128π 17. a. $\mathbf{r}(t) = \langle t, t \rangle, 1 \leq t \leq 10$

b. $|\mathbf{r}'(t)| = \sqrt{2}$ c. $\frac{\sqrt{2}}{2} \ln 10$ 19. a. $\mathbf{r}(t) = \langle 2 \cos t, 4 \sin t \rangle,$

$0 \leq t \leq \pi$ b. $|\mathbf{r}'(t)| = 2\sqrt{\sin^2 t + 4 \cos^2 t}$

c. $-8 \left[2 + \frac{1}{\sqrt{3}} \ln(2 + \sqrt{3}) \right] = -22.0828$ 21. $\frac{15}{2}$ 23. 0 25. 0

27. $\frac{3\sqrt{14}}{2}$ 29. $-2\pi^2\sqrt{10}$ 31. $\sqrt{101}$ 33. $\frac{17}{2}$ 35. 49

37. $\frac{3}{4\sqrt{10}}$ 39. 0 41. 16 43. 0 45. $\frac{3\sqrt{3}}{10}$ 47. 0

49. a. Negative b. -4π 51. a. True b. True c. True

d. True 53. a. Both paths require the same work: $W = 28,200$.

b. Both curves require the same work: $W = 28,200$.

55. a. $\frac{5\sqrt{5} - 1}{12}$ b. $\frac{5\sqrt{5} - 1}{12}$ c. The results are identical.

57. Hint: $\int_C \mathbf{F} \cdot \mathbf{T} ds = \pi r^2(c - b)$

59. Hint: $\int_C \mathbf{F} \cdot \mathbf{n} ds = \pi r^2(a + d)$ 61. The work equals zero for

all three paths. 63. 409.5 65. a. $\ln a$ b. No c. $\frac{1}{6} \left(1 - \frac{1}{a^2} \right)$

d. Yes e. $W = \frac{3^{1-p/2}}{2-p} (a^{2-p} - 1)$ for $p \neq 2$; otherwise $W = \ln a$.

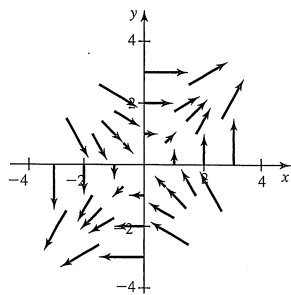
f. $p > 2$ 67. ab

Section 15.3 Exercises, pp. 984–986

1. A simple curve has no self-intersections; the initial and terminal points of a closed curve are identical. 3. Test for equality of partial derivatives as given in Theorem 14.3. 5. Integrate f with respect to x and make the constant of integration a function of y to obtain $\varphi = \int f dx + h(y)$; finally set $\frac{\partial \varphi}{\partial y} = g$ in order to determine h .
7. The integral must be zero. 9. Conservative 11. Conservative
13. Conservative 15. $\varphi(x, y) = \frac{1}{2}(x^2 + y^2)$ 17. Not conservative
19. $\varphi(x, y) = \sqrt{x^2 + y^2}$ 21. $\varphi(x, y, z) = xz + y$
23. $\varphi(x, y, z) = xy + yz + zx$ 25. $\varphi(x, y) = \sqrt{x^2 + y^2 + z^2}$
27. 0 29. 4 31. 2 33. 0 35. 0 37. 0 39. a. False
- b. True c. True d. True 41. $-\frac{1}{2}$ 43. 0 45. 10 47. 25
49. C_1 negative, C_2 positive 53. a. Compare partial derivatives.
- b. $\varphi(x, y, z) = \frac{GMm}{\sqrt{x^2 + y^2 + z^2}} = \frac{GMm}{|r|}$
- c. $\varphi(B) - \varphi(A) = GMm\left(\frac{1}{r_2} - \frac{1}{r_1}\right)$. d. No
55. a. $\frac{\partial}{\partial y} \left[\frac{-y}{(x^2 + y^2)^{p/2}} \right] = \frac{-x^2 + (p-1)y^2}{(x^2 + y^2)^{1+p/2}}$ and $\frac{\partial}{\partial x} \left[\frac{x}{(x^2 + y^2)^{p/2}} \right] = \frac{-(p-1)x^2 + y^2}{(x^2 + y^2)^{1+p/2}}$
- b. The two partial derivatives in (a) are equal if $p = 2$.
- c. $\varphi(x, y) = \tan^{-1}(y/x)$ 59. $\varphi(x, y) = \frac{1}{2}(x^2 + y^2)$
61. $\varphi(x, y) = \frac{1}{2}(x^4 + x^2y^2 + y^4)$

Section 15.4 Exercises, pp. 996–998

1. In both forms the integral of a derivative is computed from boundary data. 3. y^2 5. Area = $\frac{1}{2} \oint_C (x dy - y dx)$, where C encloses the region 7. The integral in the flux form of Green's Theorem vanishes. 9. $\mathbf{F} = \langle y, x \rangle$



11. a. 0 b. Both integrals are zero. c. Yes 13. a. -4
- b. Both integrals equal -8. c. No 15. a. 0 b. Both integrals are zero. c. Yes 17. 25π 19. 16π 21. 32 23. a. 2
- b. Both integrals equal 8π c. No 25. a. 0 b. Both integrals equal zero. c. Yes 27. a. 0 b. Both integrals equal zero.
- c. Yes 29. 6; not source free 31. $\frac{8}{3}$; not source free
33. $\frac{23\pi^2}{2} - 48$; not conservative 35. a. The circulation is zero.
- b. the outward flux equals 3π . 37. a. The circulation is zero.
- b. The outward flux equals $-\frac{15\pi}{2}$. 39. a. True b. False
- c. True 41. The circulation is zero; the outward flux equals 2π .
43. The circulation is 5702.4; the outward flux equals zero.

45. Note: $\frac{\partial f}{\partial y} = 0 = \frac{\partial g}{\partial x}$ 47. The integral becomes $\iint_R 2 dA$.

49. a. $f_x = g_y = 0$ b. $\psi(x, y) = -2x + 4y$
51. a. $f_x = e^{-x} \sin y = -g_y$ b. $\psi(x, y) = e^{-x} \cos y$
53. a. Hint: $f_x = e^x \cos y, f_y = -e^x \sin y, g_x = -e^x \sin y, g_y = -e^x \cos y$ b. Both integrals equal 0.
- c. $\varphi(x, y) = e^x \cos y, \psi(x, y) = e^x \sin y$

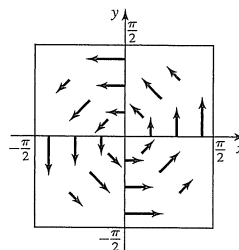
55. a. Hint: $f_x = \frac{-y}{x^2 + y^2}, f_y = \frac{x}{x^2 + y^2},$

$g_x = \frac{x}{x^2 + y^2}, g_y = \frac{y}{x^2 + y^2}$. b. Both integrals equal 0.

c. $\varphi(x, y) = x \arctan\left(\frac{y}{x}\right) + \frac{y}{2} \ln(x^2 + y^2) - y,$

$\psi(x, y) = y \arctan\left(\frac{y}{x}\right) - \frac{x}{2} \ln(x^2 + y^2) + x$

57. a.

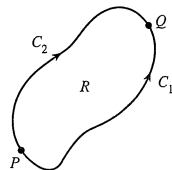


$\mathbf{F} = \langle -4 \cos x \sin y, 4 \sin x \cos y \rangle$ b. Yes, the divergence equals zero. c. No, the two-dimensional curl equals $8 \cos x \cos y$.

d. The total flux across the boundary is zero. e. The total

circulation along C is 32. 59. $\mathbf{F} = \left\langle \frac{f(x)}{d-c}, 0 \right\rangle$ for the rectangle $[a, b] \times [c, d]$. 61. c. The vector field is undefined at the origin.

63.



Basic ideas: Let C_1 and C_2 be two smooth simple curves from A to B .

$$\int_{C_1} \mathbf{F} \cdot \mathbf{n} ds - \int_{C_2} \mathbf{F} \cdot \mathbf{n} ds = \oint_C \mathbf{F} \cdot \mathbf{n} ds = \iint_R \operatorname{div} \mathbf{F} dA = 0$$

$$\text{and } \int_{C_1} \mathbf{F} \cdot \mathbf{n} ds = \int_{C_1} \psi_x dx + \psi_y dy = \int_{C_1} d\psi = \psi(B) - \psi(A)$$

65. Use $\nabla \varphi \cdot \nabla \psi = \langle f, g \rangle \cdot \langle -g, f \rangle = 0$

Section 15.5 Exercises, pp. 1006–1009

1. Compute $f_x + g_y + h_z$. 3. There are no sources or sinks.
5. It indicates the axis and the angular speed of the circulation at a point. 7. Zero 9. 3 11. 0 13. $2(x + y + z)$
15. $\frac{x^2 + y^2 + 3}{(1 + x^2 + y^2)^2}$ 17. $\frac{1}{|r|^2}$ 19. $\frac{-1}{|r|^4}$ 21. a. Positive for both points b. $\operatorname{div} \mathbf{F} = 2$ c. Outward everywhere d. Positive
23. a. $\operatorname{curl} \mathbf{F} = 2\mathbf{i}$ b. $|\operatorname{curl} \mathbf{F}| = 2$
25. a. $\operatorname{curl} \mathbf{F} = 2\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$ b. $|\operatorname{curl} \mathbf{F}| = 2\sqrt{3}$
27. $3y\mathbf{k}$ 29. $-4z\mathbf{j}$ 31. 0 33. 0 35. Follows from partial differentiation of $\frac{1}{(x^2 + y^2 + z^2)^{3/2}}$

37. Combine Exercise 36 with Theorem 14.8. 39. a. False
 b. False c. False d. False e. False 41. a. No b. No
 c. Yes, scalar function d. No e. No f. No g. Yes, vector
 field h. No i. Yes, vector field 43. Compute an explicit
 expression for $\mathbf{a} \times \mathbf{r}$ and then take the required partial derivatives.
 45. $\text{div } \mathbf{F} = 6$ at $(1, 1, 1)$, $(1, -1, -1)$, $(-1, 1, 1)$, and $(-1, -1, -1)$.
 47. $\mathbf{n} = \langle a, b, 2a + b \rangle$, where a and b are real numbers
 49. $\mathbf{F} = \frac{1}{2}(y^2 + z^2)\mathbf{i}$ 51. a. The wheel does not spin.
 b. Clockwise, looking in the positive y -direction
 c. The wheel does not spin. 53. $\omega = \frac{10}{\sqrt{3}}$, or $\frac{5}{\sqrt{3}\pi} = 0.9189$

revolutions per unit time.

55. $\mathbf{F} = -200ke^{-x^2+y^2+z^2}(-x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$
 $\nabla \cdot \mathbf{F} = -200k(1 + 2(x^2 + y^2 + z^2))e^{-x^2+y^2+z^2}$

57. a. $\mathbf{F} = \frac{GMm\mathbf{r}}{|\mathbf{r}|^3}$ b. See Theorem 14.9.

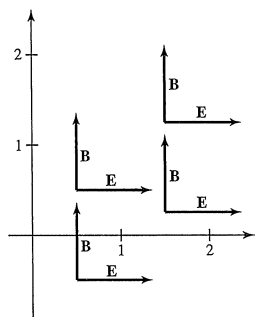
59.
$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

$$\rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right)$$

$$\rho \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right)$$

61. a. Use $\nabla \times \mathbf{B} = -Ak \cos(kz - \omega t)\mathbf{i}$ and

$\frac{\partial \mathbf{E}}{\partial t} = -A\omega \cos(kz - \omega t)\mathbf{i}$ b.



Section 15.6 Exercises, pp. 1022–1025

1. $\mathbf{r}(u, v) = \langle a \cos u, a \sin u, v \rangle$, $0 \leq u \leq 2\pi$, $0 \leq v \leq h$
 3. $\mathbf{r}(u, v) = \langle a \sin u \cos v, a \sin u \sin v, a \cos u \rangle$, $0 \leq u \leq \pi$,
 $0 \leq v \leq 2\pi$ 5. Use the parameterization from Problem 3 and compute

$$\int_0^\pi \int_0^{2\pi} f(a \sin u \cos v, a \sin u \sin v, a \cos u) a^2 \sin u \, dv \, du$$

 7. Use the parametrization from Problem 3 and compute

$$\int_0^\pi \int_0^{2\pi} a^2 \sin u (f \sin u \cos v + g \sin u \sin v + h \cos u) \, dv \, du$$

 9. The normal vectors point outward. 11. $\langle u, v, \frac{1}{3}(16 - 2u + 4v) \rangle$,
 $|u| < \infty$, $|v| < \infty$ 13. $\langle v \cos u, v \sin u, v \rangle$, $0 \leq u \leq 2\pi$,
 $2 \leq v \leq 8$ 15. $\langle 3 \cos u, 3 \sin u, v \rangle$, $0 \leq u \leq \frac{\pi}{2}$, $0 \leq v \leq 3$
 17. A plane above the square $[1, 3] \times [2, 4]$ 19. Part of the upper
 half of the cone $z^2 = 16x^2 + 16y^2$ of height 12 and radius 3 (with
 $y \geq 0$) 21. 28π 23. $16\sqrt{3}$ 25. $\pi r \sqrt{r^2 + h^2}$ 27. 1728π

29. 0 31. $4\pi\sqrt{5}$ 33. $8\sqrt{17} + 2 \ln(\sqrt{17} + 4) = 37.1743$
 35. $\frac{2\sqrt{3}}{3}$ 37. $\frac{1250\pi}{3}$ 39. $\frac{1}{48}(e - e^{-5} - e^{-7} + e^{-13})$ 41. $\frac{1}{4\pi}$
 43. -8 45. 0 47. $4\pi a$ 49. a. True b. False c. True d. True
 51. $8\pi(4\sqrt{17} + \ln(\sqrt{17} + 4))$ 53. $8\pi a$ 55. a. 8 b. $4\pi - 8$
 57. a. 0 b. 0 The flow is always tangential to the surface (radial
 flow). 59. $2\pi ah$ 61. $-400\left(e - \frac{1}{e}\right)^2$ 63. $8\pi a$
 65. a. $4\pi(b^3 - a^3)$ b. The net flux is zero. 67. $(0, 0, \frac{2}{3}h)$
 69. $(0, 0, \frac{7}{6})$ 73. Flux = $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_A dx \, dy$

Section 15.7 Exercises, pp. 1032–1034

1. The integral measures the circulation along the closed curve C .
 3. The circulation along a closed curve can be calculated by integrat-
 ing the dot product of the curl and the normal vector on an enclosed
 surface. 5. -2π for both integrals. 7. Both integrals are zero.
 9. -18π for both integrals. 11. -24π 13. $-\frac{128}{3}$ 15. 15π
 17. 0 19. -32π 21. $\nabla \times \mathbf{v} = \langle 1, 0, 0 \rangle$; a paddle wheel with its
 axis aligned with the x -axis will spin with maximum angular speed
 counterclockwise (looking in the negative x -direction) at all points.
 23. $\nabla \times \mathbf{v} = \langle 0, -2, 0 \rangle$; a paddle wheel with its axis aligned with the
 y -axis will spin with maximum angular speed clockwise (looking in the
 negative y -direction) at all points. 25. a. False b. False c. True
 d. True 27. The circulation is zero. 29. The circulation is zero.
 31. 2π 33. $\pi(\cos \varphi - \sin \varphi)$, maximum for $\varphi = 0$ 35. The circula-
 tion is 48π ; it depends on the radius of the circle but not on the center.
 37. a. The normal vectors point toward the z -axis on the curved sur-
 face of S and in the direction of $\langle 0, 1, 0 \rangle$ on the flat surface of S .
 b. 2π c. 2π 39. The integral is π for all a . 41. The integral is zero
 for (a) and (b). 43. b. 2π for any circle of radius r centered at the
 origin. c. $\mathbf{F}$ is not differentiable along the z -axis. 45. Apply the
 Chain Rule. 47. $\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_R \frac{\partial h}{\partial y} - \frac{\partial g}{\partial z} \, dA$

Section 15.8 Exercises, pp. 1043–1046

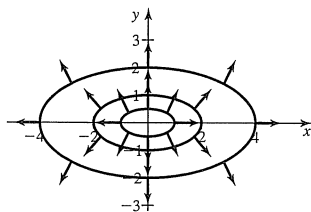
1. The surface integral measures the flow across the boundary.
 3. The flux across the boundary equals the cumulative expansion
 or contraction of the vector field inside the region. 5. 32π
 7. The outward fluxes are equal to each other. 9. Both integrals
 equal 96π . 11. Both integrals are zero. 13. The net flux is zero.
 15. The net flux is zero. 17. $16\sqrt{6}\pi$ 19. $\frac{2}{3}$ 21. $-\frac{128}{3}\pi$
 23. 24π 25. -224π 27. 12π 29. 20 31. a. False
 b. False c. True 33. 0 35. $\frac{3}{2}$ 37. b. The net flux between the
 two spheres is $4\pi(a^2 - \varepsilon^2)$. 39. b. Use $\nabla \cdot \mathbf{E} = 0$. c. The flux
 across S is the sum of the contributions from the individual charges.
 d. For an arbitrary volume, we find

$$\frac{1}{\varepsilon_0} \iiint_D q(x, y, z) \, dV = \iint_S \mathbf{E} \cdot \mathbf{n} \, dS = \iiint_D \nabla \cdot \mathbf{E} \, dV$$

 e. Use $\nabla^2 \varphi = \nabla \cdot \nabla \varphi$. 41. 0 43. $-(1 - e^{-1})$ 45. $800\pi a^3 e^{-a^2}$

Chapter 15 Review Exercises, pp. 1046–1049

1. a. False b. True c. False d. False e. True
3.



5. $-\frac{\mathbf{r}}{|\mathbf{r}|^3}$ 7. a. $\mathbf{n} = \frac{1}{2}\langle x, y \rangle$ b. 0 c. $\frac{1}{2}$ 9. $\frac{\sqrt{46}}{4}(e^{6(\ln 8)^2} - 1)$

11. Both integrals are zero. 13. 0 15. The circulation is -4π ; the outward flux is zero. 17. The circulation is zero; the outward flux is

2π . 19. $\frac{4v_0 L^3}{3}$ 21. $\varphi(x, y, z) = xy + yz^2$ 23. $\varphi(x, y, z) = xye^z$

25. 0 for both methods 27. a. $-\pi$ b. $\mathbf{F}$ is not conservative.

29. 0 31. $\frac{20}{3}$ 33. 8π 35. The circulation is zero; the outward flux equals 2π . 37. a. $b = c$ b. $a + d = 0$ c. $a + d = 0$ and $b = c$

39. $\nabla \cdot \mathbf{F} = 4\sqrt{x^2 + y^2 + z^2} = 4|r|$, $\nabla \times \mathbf{F} = \mathbf{0}$, $\nabla \cdot \mathbf{F} \neq 0$

41. $\nabla \cdot \mathbf{F} = 2y + 12xz^2$, $\nabla \times \mathbf{F} = \mathbf{0}$, $\nabla \cdot \mathbf{F} \neq 0$ 43. a. -1 and 0

b. $\mathbf{n} = \frac{1}{\sqrt{3}}\langle -1, 1, 1 \rangle$ 45. 18π 47. $4\sqrt{3}$ 49. $\frac{8\sqrt{3}}{3}$

51. 8π 53. $4\pi a^2$ 55. a. Use $x = y = 0$ to confirm the highest point; use $z = 0$ to confirm the base. b. The hemisphere S has the greater surface area— $2\pi a^2$ for S versus $\frac{5\sqrt{5}-1}{6}\pi a^2$ for T .

57. 0 59. 99π 61. 0 63. $\frac{972}{5}\pi$ 65. $\frac{124}{5}\pi$ 67. $\frac{32}{3}$

APPENDIX A

Exercises, pp. 1056–1057

1. The set of real numbers greater than -4 and less than or equal to 10 ; $(-4, 10]$;

3. $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$ 5. $2x - 4 \geq 3$ or $2x - 4 \leq -3$

7. Take the square root of the sum of the squares of the differences of the x - and y -coordinates. 9. $y = \sqrt{36 - x^2}$

11. $m = \frac{y+2}{x-4}$ or $y = m(x-4) - 2$ 13. They are equal.

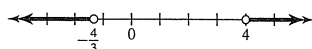
15. 4 17. $4uv$ 19. $\frac{-h}{x(x+h)}$ 21. $(y - y^{-1})(y + y^{-1})$

23. $u = \pm\sqrt{2}, \pm 3$ 25. $3x^2 + 3xh + h^2$

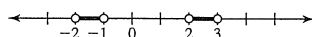
27. $(1, 5)$

29. $(-\infty, 4] \cup [5, 6)$

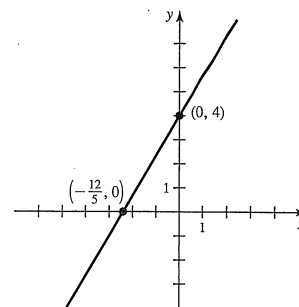
31. $\{x: x < -4/3 \text{ or } x > 4\}; (-\infty, -4/3) \cup (4, \infty)$



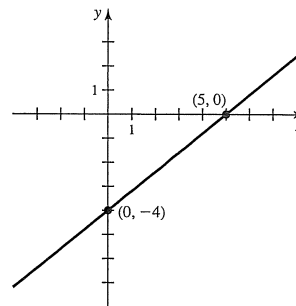
33. $\{x: -2 < x < -1 \text{ or } 2 < x < 3\}; (-2, -1) \cup (2, 3)$



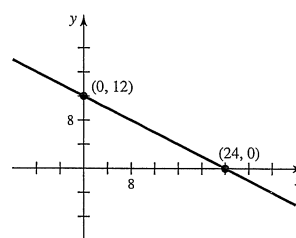
35. $y = 2 - \sqrt{9 - (x+1)^2}$ 37. $y = \frac{5}{3}x + 4$



39. $y = \frac{4}{5}x - 4$



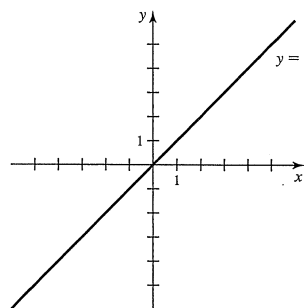
41. $x + 2y = 24$



43. $y = \frac{1}{3}x - 7$ 45. a. False b. True c. False d. False

e. False f. True g. False 47. $\{x: |x - 1| \geq 3\}$

49.



APPENDIX C

Exercises, pp. 1069–1070

1. $R, \{x: x \neq 0\}, R, \{x: x \neq 0\}$ 3. $\frac{3x \sinh 3x - 2 \cosh 3x}{x^3}$

5. $y'(x) = \frac{\operatorname{sech}(xy) - y}{x}$ 7. $64/81$

9. Sample proofs: $\frac{d}{dx}(\coth x) = \frac{d}{dx}\left(\frac{\cosh x}{\sinh x}\right)$

$$= \frac{\sinh^2 x - \cosh^2 x}{\sinh^2 x} = -\frac{1}{\sinh^2 x} = -\operatorname{csch}^2 x$$

$$\frac{d}{dx}(\operatorname{sech} x) = \frac{d}{dx}(\cosh x)^{-1} = -(\cosh x)^{-2} \sinh x$$

$$= -\frac{1}{\cosh x} \frac{\sinh x}{\cosh x} = -\operatorname{sech} x \tanh x$$

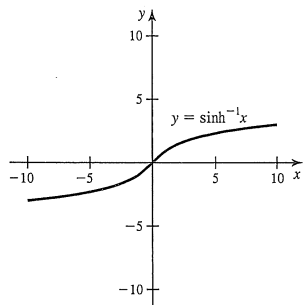
11. $\frac{3 \operatorname{sech}^2 3x}{2\sqrt{\tanh 3x}}$ 13. $-\coth x + C$ 15. $\frac{1}{5} \ln|\sinh 5x| + C$

17. $2\ln\left(\frac{\cosh 2}{\cosh 1}\right) \approx 1.78$ 19. π

21. $L = \int_0^a \sqrt{1 + \sinh^2 x} dx = \int_0^a \cosh x dx = \sinh a$

23. b. Domain = $\mathbf{R}$; range = $\mathbf{R}$

c.



d. $y = \sinh^{-1} x \Rightarrow x = \sinh y$. Differentiating with respect to x gives

$$1 = (\cosh y)y'(x) \quad \text{or} \quad y'(x) = \frac{1}{\cosh y}$$

Now $\cosh^2 y - \sinh^2 y = 1$ implies that $\cosh y = \sqrt{1 + \sinh^2 y} = \sqrt{1 + x^2}$. Therefore,

$$y'(x) = \frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1 + x^2}} \quad \text{25. } \ln(\sqrt{x^2 + 9} + x) + C$$

27. a. $a = 2$ b. $L = 2\ln(2 + \sqrt{3}) \approx 2.63$

c.

